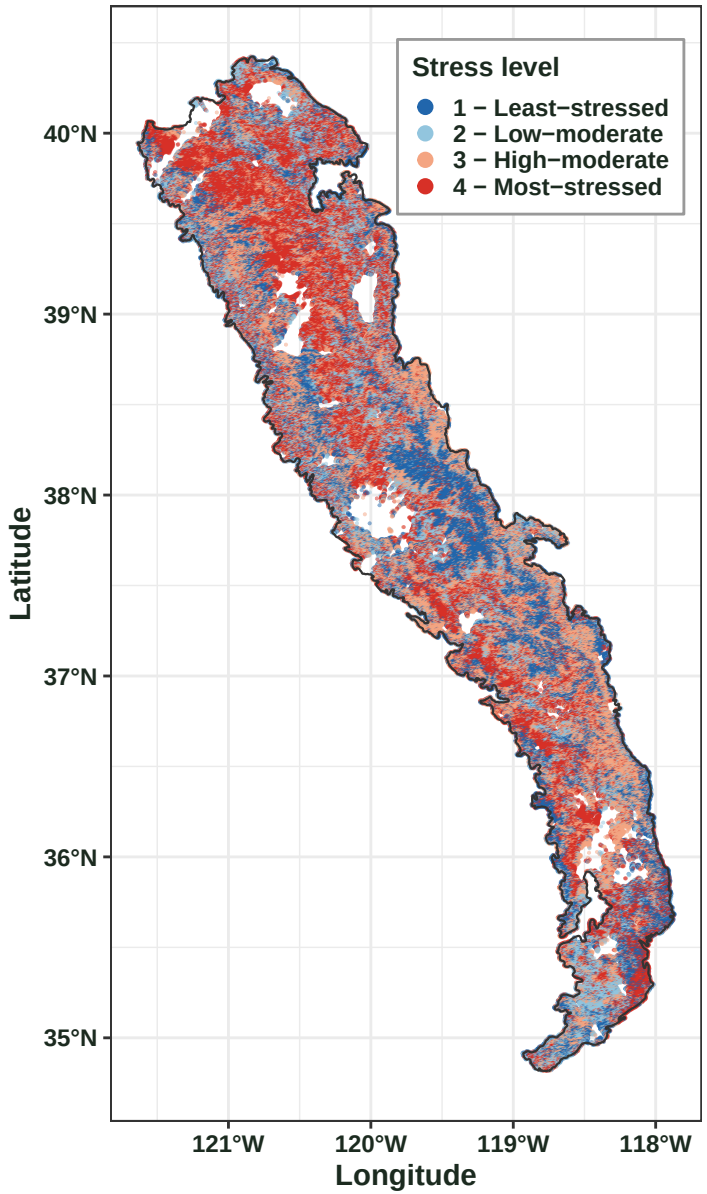
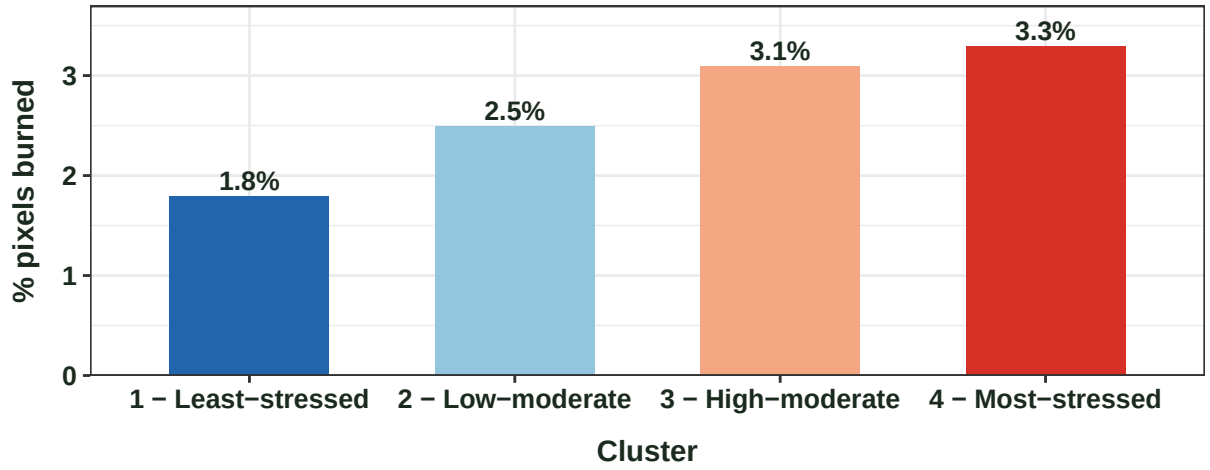
# NDVI+NDWI−combined: Cluster Map + Real−World Validation Summary

## Cluster Map



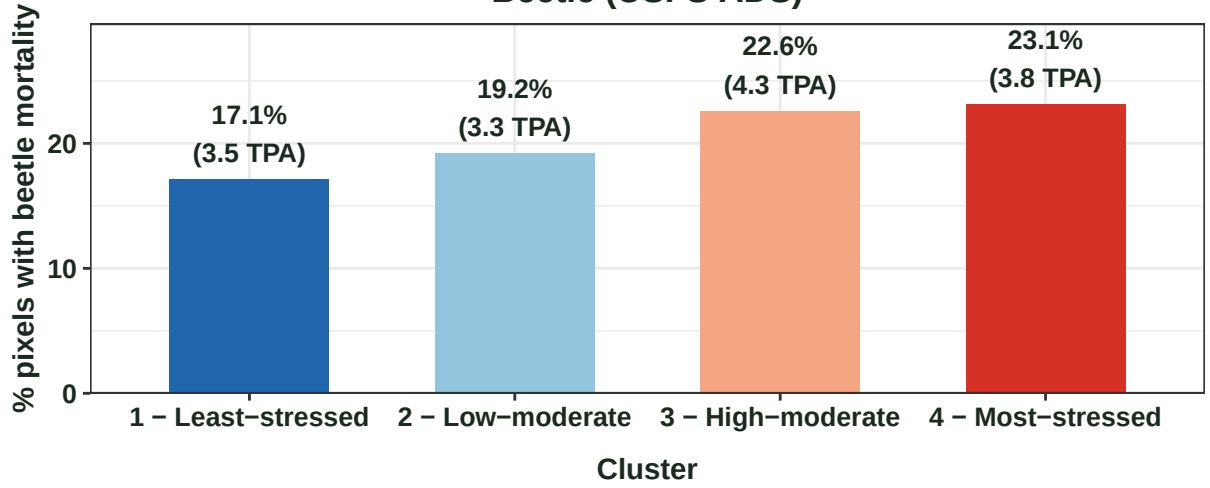

White gaps: pixels burned (MTBS moderate+ severity) at any point 2000−2014, excluded before clustering

## Fire (MTBS)




**Relative risk (Cluster 4 vs 1): 1.8×**

## Beetle (USFS ADS)




**Relative risk (Cluster 4 vs 1): 1.4×**